

Almost-contractive linear recovery with exact volume scaling

MathlibAnnex.PluckerRecovery.nonempty_linearCertificate_of_pluckerBodies_eq

theorem

Recovers one bijective linear factor with a norm estimate and exact volume scaling.

Statement

Let $n = m + 1 > 0$ and $V = \mathbb{R}^n$ with the reference sup norm and standard Lebesgue measure λ . The continuous norms p, q have fixed positive comparison constants

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty, \quad a_q \|x\|_\infty \leq q(x) \leq b_q \|x\|_\infty.$$

Set $K_p = \{x : p(x) \leq 1\}$, $K_q = \{x : q(x) \leq 1\}$ and $v_p = \lambda(K_p)$, $v_q = \lambda(K_q)$, as positive finite real numbers. For a real linear map $A : V \rightarrow \mathbb{R}^N$, let $A_{ij} = (Ae_j)_i$ be its matrix in the standard bases. The minor-coordinate space has one coordinate for each increasing n -tuple I of distinct rows from $\{1, \dots, N\}$, and

$$\Pi_n(A)_I = \det(A_{I,*}).$$

Define the signed generator set and its real convex hull by

$$\mathcal{G}_p^N = \left\{ \sigma v_p \Pi_n(A) : \begin{array}{l} A : V \rightarrow \mathbb{R}^N \text{ is real linear, } \sigma \in \{1, -1\}, \\ \|Ax\|_\infty \leq p(x) \quad (x \in V) \end{array} \right\},$$

$$\mathcal{P}_p^N = \text{conv}_{\mathbb{R}} \mathcal{G}_p^N.$$

Define $\mathcal{G}_q^N, \mathcal{P}_q^N$ in the same way using q, v_q . The generators are vectors of minors, not linear maps. Suppose $\varepsilon > 0$ and $\mathcal{P}_p^N = \mathcal{P}_q^N$ for every N . There exist $r \in \mathbb{N}$, real linear maps $B_p, B_q : V \rightarrow \mathbb{R}^{n+r}$, a real linear bijection $L : V \rightarrow V$ with

$$B_q L = B_p, \quad \text{range } B_p = \text{range } B_q,$$

$$\forall x, (1 - \varepsilon) q(Lx) \leq p(x), \quad v_p |\det L| = v_q.$$

Assumptions

The coordinate dimension is positive. The inputs are the norms p, q with the stated positive comparison bounds, $\varepsilon > 0$, and equality $\mathcal{P}_p^N = \mathcal{P}_q^N$ for every nonnegative N . The nonzero leading minors used in the proof are obtained during the construction, not assumed.

Conclusion

The same bijection L factors the selected representatives and satisfies both the norm estimate and the exact volume equation. No upper restriction $\varepsilon < 1$ is required.

Proof route

Anchor the common row point at a nonzero leading coordinate, recover its chart factor, and compute the estimate and volume identity for that factor.

▼ Proof steps

1. **Anchor the same representatives.** Let $D_q > 0$ be the maximum absolute determinant of a family of n real linear functionals dominated by q , and choose target satellite parameters with $\eta = D_q/2$. The absolute maximizing target base has

$$|\det M_{F_q}| \geq D_q - \eta > 0.$$

Thus every target support maximizer has nonzero leading minor coordinate, by the leading-coordinate lemma. anchored common row selection supplies one z and the same contractive B_p, B_q , with signs $\sigma_p, \sigma_q \in \{1, -1\}$, the target condition $1 - \varepsilon < \|B_q x\|_\infty$ whenever $q(x) = 1$, and

$$z = \sigma_p v_p \Pi_n(B_p) = \sigma_q v_q \Pi_n(B_q), \quad z_{I_0} \neq 0.$$

Here $I_0 = (1, \dots, n)$ selects the first n rows. Let $S_p = (B_p)_{I_0, *}$ and $S_q = (B_q)_{I_0, *}$ be the corresponding square matrices. Both are nonsingular because their determinants give the nonzero coordinate z_{I_0} , up to nonzero signs and volumes. Put $\tau = \sigma_p \sigma_q$. Each coordinate gives

$$v_p \det(B_p)_{I, *} = \tau v_q \det(B_q)_{I, *}.$$

This is the weighted minor relation. Its nonzero anchor uses the determinant gap, including when $\varepsilon \geq 1$.

2. **Recover the factor, row by row.** Put

$$c = \tau v_q / v_p \neq 0, \quad L = S_q^{-1} S_p.$$

The maximal minors of B_p are c times those of B_q . The same proportionality holds for any ordered row selection: sorting contributes the same permutation sign on both sides, and a repeated row makes both determinants zero. This is the passage from increasing to ordered minors.

Fix corresponding row vectors u_p of B_p and u_q of B_q . Applying the proportionality to the leading block and to its i th-row replacement gives

$$\det S_p = c \det S_q, \quad \det S_p[i \leftarrow u_p] = c \det S_q[i \leftarrow u_q].$$

Row Cramer gives each coordinate of the two normalized rows. Substituting both proportionalities, with $c, \det S_p, \det S_q \neq 0$, gives

$$\begin{aligned}
(u_p S_p^{-1})_i &= \frac{\det S_p[i \leftarrow u_p]}{\det S_p} \\
&= \frac{c \det S_q[i \leftarrow u_q]}{c \det S_q} \\
&= \frac{\det S_q[i \leftarrow u_q]}{\det S_q} \\
&= (u_q S_q^{-1})_i.
\end{aligned}$$

This is the source's row-coordinate calculation for these two matrices and this fixed row. Equality in every coordinate now yields

$$\begin{aligned}
u_p &= (u_p S_p^{-1}) S_p \\
&= (u_q S_q^{-1}) S_p \\
&= u_q (S_q^{-1} S_p) = u_q L.
\end{aligned}$$

Thus $B_q L = B_p$ on all rows, as recorded by the chart factor theorem. Both leading blocks are nonsingular, so this same L is invertible by chart factor invertibility. Since L is onto,

$$B_p(V) = B_q(L(V)) = B_q(V).$$

For fixed B_p, B_q the factor is unique: $B_q L' = B_p$ implies $S_q L' = S_p$, hence $L' = S_q^{-1} S_p = L$. This uniqueness concerns the fixed representatives, not all possible choices of them.

3. **Evaluate the same factor in the norm estimate.** If $x \neq 0$, injectivity gives $Lx \neq 0$. Homogeneity of the target lower condition at Lx gives

$$(1 - \varepsilon)q(Lx) < \|B_q Lx\|_\infty = \|B_p x\|_\infty \leq p(x).$$

If $x = 0$, both sides are zero. The resulting universal non-strict estimate is the factor model bound. There is no division by $1 - \varepsilon$.

4. **Compute its determinant and volume.** The leading minor of $B_q L = B_p$ gives $\det S_p = \det S_q \det L$. Substitute into the weighted leading-minor relation:

$$v_p \det S_q \det L = \tau v_q \det S_q.$$

Cancel $\det S_q \neq 0$ to obtain $v_p \det L = \tau v_q$. Taking absolute values, using $v_p, v_q > 0$ and $|\tau| = 1$, gives

$$v_p |\det L| = v_q.$$

The signed determinant identity and absolute volume identity use this same factor.

Factor uniqueness in the proof is for fixed anchored representatives; the public certificate only asserts existence. It does not publicly store the intermediate maps' two contractivity fields. Different common-point selections can differ.

Main citations

- Exact declaration and source

- the leading-coordinate lemma
- anchored common row selection
- the weighted minor relation
- the passage from increasing to ordered minors
- row-coordinate calculation
- the chart factor theorem
- chart factor invertibility
- the factor model bound
- signed determinant identity
- absolute volume identity
- The twelve-field linear certificate
- the signed generator set
- its real convex hull

Lean source signature (exact)

```
theorem nonempty_linearCertificate_of_pluckerBodies_eq
  {m : ℕ} (MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ)) {ε : ℝ} (hε : 0 < ε)
  (hBodies : ∀ N : ℕ, MathlibAnnex.PluckerBody.body MX N = MathlibAnnex.PluckerBody.body
MY N) :
  Nonempty (LinearCertificate MX MY ε)
```

In the source	Mathematical meaning
$\text{Fin } (m + 1) \rightarrow \mathbb{R}; \rightarrow L[\mathbb{R}]$	The reference $V = \mathbb{R}^n, n = m + 1 > 0$, with sup norm; arrows mean continuous real linear maps. Source indices $0, \dots, n - 1$ correspond to formula indices $1, \dots, n$.
$\text{MX MY : EquivalentSeminorm } (\text{Fin } (m + 1) \rightarrow \mathbb{R});$ $\text{MX.p};$ MY.p	The two continuous norms p, q and their positive comparison constants. The applications $\text{MX.p } x$ and $\text{MY.p } x$ mean $p(x), q(x)$.
$\text{MX.closedUnitBall}; \text{MY.closedUnitBall};$ $\text{MX.closedUnitBallVolume};$ $\text{MY.closedUnitBallVolume}$	The sets K_p, K_q and their real Lebesgue volumes v_p, v_q . These finite nonnegative measures are read as real numbers.
$\text{hBodies : } \forall N : \mathbb{N}, \text{PluckerBody.body MX N} =$ $\text{PluckerBody.body MY}$ N	For every nonnegative N , the real convex hulls $\mathcal{P}_p^N$ and $\mathcal{P}_q^N$ are equal. The source body is the convex hull of the vectors $\pm v_p \Pi_n(A)$, where $A : V \rightarrow \mathbb{R}^N$ is real linear and $\ Ax\ _\infty \leq p(x)$ for every x . The target body uses q, v_q in place of p, v_p .
$\text{hε : } 0 < \varepsilon; \text{Nonempty } (\text{LinearCertificate MX MY}$ $\varepsilon)$	Positive ε and existence of one record carrying all twelve fields below.

The inputs are the two norm models, hε and hBodies . The conclusion $\text{Nonempty } (\text{LinearCertificate MX MY } \varepsilon)$ asserts that all twelve fields below can be filled for the same chosen objects.

Related structure, shown separately: The twelve-field linear certificate.

```

structure LinearCertificate {m : ℕ}
  (MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ)) (ε : ℝ) where
  q : ℕ
  sourceMap : (Fin (m + 1) → ℝ) →L[ℝ] (Fin ((m + 1) + q) → ℝ)
  targetMap : (Fin (m + 1) → ℝ) →L[ℝ] (Fin ((m + 1) + q) → ℝ)
  linearMap : (Fin (m + 1) → ℝ) →L[ℝ] (Fin (m + 1) → ℝ)
  factor : targetMap.comp linearMap = sourceMap
  range_sourceMap_eq_range_targetMap : Set.range sourceMap = Set.range targetMap
  injective : Function.Injective linearMap
  surjective : Function.Surjective linearMap
  one_sub_mul_seminorm_linearMap_le : ∀ x, (1 - ε) * MY.p (linearMap x) ≤ MX.p x
  det : ℝ
  det_eq : det = ContinuousLinearMap.det linearMap
  closedUnitBallVolume_mul_abs_det : MX.closedUnitBallVolume * |det| =
MY.closedUnitBallVolume

```

In the source	Mathematical meaning
<code>q : ℕ</code>	The extra coordinate dimension $r \in \mathbb{N}$, so the common codomain is $\mathbb{R}^{n+r}$. This field is not the target norm q .
<code>sourceMap</code>	The continuous real linear map $B_p : V \rightarrow \mathbb{R}^{n+r}$.
<code>targetMap</code>	The continuous real linear map $B_q : V \rightarrow \mathbb{R}^{n+r}$.
<code>linearMap</code>	The same continuous real linear factor $L : V \rightarrow V$.
<code>factor : targetMap.comp linearMap = sourceMap</code>	$B_q \circ L = B_p$, that is, $B_q(Lx) = B_p x$ for every x ; <code>.comp</code> is composition in this order.
<code>range_sourceMap_eq_range_targetMap</code>	Equality of the image sets $\{B_p x : x \in V\} = \{B_q x : x \in V\}$.
<code>injective</code>	$Lx = Ly$ implies $x = y$.
<code>surjective</code>	For each $y \in V$ there is $x \in V$ such that $Lx = y$.
<code>one_sub_mul_seminorm_linearMap_le</code>	The estimate $(1 - \varepsilon)q(Lx) \leq p(x)$ for every $x \in V$.
<code>det : ℝ</code>	The stored real value of $\det L$; its identification is recorded by the next field.
In the source	Mathematical meaning
<code>det_eq : det = ContinuousLinearMap.det linearMap</code>	The field <code>det</code> equals $\det L$. This is the determinant of the square linear factor L , not of either rectangular map.
<code>closedUnitBallVolume_mul_abs_det</code>	Using <code>det_eq</code> , this field states $v_p \det L = v_q$, for the same L .

Exact definitions: the signed generator set; its real convex hull.

Surrounding assumptions and aliases.

▼ **Exact content identity**

Declaration: `MathlibAnnex.PluckerRecovery.nonempty_linearCertificate_of_pluckerBodies_eq`

Accepted content SHA-256: `b079b84f15ab93a5cab899b7d29f497ca4248d03dadecee1748b96afb1f855e`

Accepted source guide SHA-256: `8c1e0fb1cbe47d104f9c2750a5fe7f090c99c9eff7e1827c6889a9633a0e8f76`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`